Enrollment No:.....

Supplementary Examination – March 2025

Programme: B.Tech[CSE/IT/CST/ITE/CS(AI)/CS(AIML)/CS(DS)] **Semester: 4rd Semester**

Paper Code: CIC-212 **Paper Name: Programming in Java**

Time: 1½Hrs. **Maximum Marks: 30**

Note:
- Question No. 1 is compulsory.
- Attempt any two questions from the remaining questions.
- Some questions have internal choice also.
- All questions carry equal marks.

	Question 1	Marks
1(a)	Can a Java interface have a constructor? Why or why not?	[2]
1(b)	Explain the purpose of class member modifiers public, private, protected in Java.	[2]
1(c)	Discuss the concept of method overriding in Java.	[2]
1(d)	Explain the purpose of nameless object in java.	[2]
1(e)	Explain the concept of Java applets and their typical use cases?	[2]
	Question 2	
2(a)	Explain the use of final keyword in java?	[2]
2(b)	Compare and contrast the wrapper classes for different primitive data types?	[4]
2(c)	Explain the methods provided by the StringBuffer class, such as append(), insert(), delete(), and reverse(). Provide examples demonstrating their usage.	[4]
	Question 3	
3(a)	Discuss the difference between JDK, JRE, and JVM.	[2]
3(b)	Explain the concept of "type casting" in Java. What are the differences between implicit and explicit casting?	[3]
3(b)	Provide examples to illustrate inheritance, polymorphism, encapsulation, and abstraction.	[5]
	Question 4	
4(a)	Discuss the security promises of the JVM. How does the JVM protect Java applications from various types of vulnerabilities?	[5]
4(b)	Describe how user-defined exceptions can be created in Java. Provide an example where a custom exception is used to handle specific application errors.	[5]

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